

AeroVis - Mixed Reality Wire Harness

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ABSTRACT - Overview: AeroVis aims to revolutionize the design process and maintenance in the aviation industry through the implementation of mixed reality. This is done by converting a schematic into a binary map and using the A* algorithm to optimize the pathfinding of the components in the harness.

Methods: Utilizing the Unity Engine in order to develop a Mixed Reality (MR) application. The application is loaded onto a Microsoft HoloLens2 for use in Mixed Reality. The application is developed using C# and Python inside of the Unity Engine v2022.3.45f1. Using a user-uploaded schematic, a python script will convert that image into grayscale. This grayscale image will be used to make a binary map of the schematic. This binary map will use an A* algorithm in order to route out the most efficient paths for the wire harness and display the result to the user.

Keywords - Mixed Reality, Unity Development, HoloLens2, A* Algorithm

I. INTRODUCTION

A. The Problem

The problem that AeroVis seeks to solve is a problem that every individual in any manufacturing, design, or engineering field has experienced before: the requirement to walk away from our work to look at the schematics. AeroVis seeks to solve that problem by implementing the use of Mixed Reality into the manufacturing process through the eyes of Microsoft's HoloLens2 Mixed Reality Headset. The aim of the project is to display an accurate, to scale, schematic that can be seen and interacted with while the user is still able to engage with the world around them in real time.

B. Literature Search

To begin work on AeroVis when it was pivotal to search for projects that correspond to the topic at hand and see if something similar had been done that could aid¹. Using the foundation set by the resulting sources to not only

gain a better understanding of their task but also gather insights into how others approached similar challenges and how AeroVis can build upon that pre-existing work to achieve the penultimate objective. The search results highlighted diverse applications of advanced algorithms and mixed reality (MR). They include improving helicopter safety with the city based A* Algorithm [1], enhancing industrial training with MR [2], and using MR glasses for robotic pathfinding [3]. Research also covers object-centered UIs in MR [4], interactive learning tools for aircraft engineering [5], and personalized RV design with HoloLens 2 [6]. Additionally, advancements in the A* Algorithm and its application in game pathfinding are discussed, showing improved performance in various scenarios.

C. Project Leadership

For this task the group came to the conclusion that since there were only two people it would be best to split the project in half - software and hardware. Due to the complexity of displaying the progress, design, functionality, and the lack of tangible progress; it was best to create a functional replication of the MR application to coincide with the software itself. Ryan would take leadership of the software side of the project due to his background in C# and knowledge of Unity, and Rudy would lead the physical development to demonstrate the application. This provided the group with a finished software product, while also having a physical demonstration of what AeroVis did without requiring somebody to put the headset on and load up the application.

D. Personality Tests

In order to build team chemistry from the get-go, it was decided to see how the group's personality types aligned. Using Carl Jung's and Isabel Briggs Myers' typology, it was concluded that Ryan resulted in a typology of INFJ (Introverted, iNtuitive, Feeling, Judging); and Rudy had resulted in a type of ESTJ (Extroverted, Sensing,

1. Literature search was performed on September 5, 2024 in correspondence with Sacred Heart University's Librarian Kristina D'Agostino

Thinking, Judging). These results showcased how as a group the personality difference can actually compliment the other. This promoted covering a large spectrum of the personality typology and interacting with as large of an audience as possible.

E. Complex Engineering Problem

Currently, the methods used to construct wire harnesses in the aircraft industry can be very resource intensive due to the pure size of the blueprints. This not only requires much more time swapping out the schematics, but requires much more physical storage and dedicated tracking system for the different schematics. AeroVis seeks to eliminate these resources by allowing for the user to use schematics from the memory of the HoloLens and display them on a nearby surface. This would allow for the user to put the headset on, select any schematic needed, and get to work, eliminating the need for a potential schematic swap or changing workstations. AeroVis also will compute the most efficient path for a wire to travel, vastly lowering the time spent actually preparing the schematics for the production line.

F. Engineering Solution

In order to plot out the potential directions the project can go, our project began with brainstorming solutions to reach the goal of our project. By creating a box diagram displaying what features would be ideal in the implementation of the application. For the project this year one of the main areas of interest is one centralized user-interface (UI). This would make navigation and operation of the application itself more efficient, and make it look more presentable; regardless of the finished project. Since the project previously was without a universal solution to the problems at hand, any sort of foundation that will be set in place now will be set firmly. For the project going forward, rather than building off of what was previously completed, it would be better to use their work and their failures to determine points of most significance. Analyzing these failures, along with the successes alongside them, will result in an outline of the most important topics to tackle and take on early. Most importantly, implementation of a central interface that would guide the user around the application and streamline the schematic process would make the project fit better as an application. It would also be beneficial to potentially have calibration settings for all potential users saved and be able to be called back for accessibility purposes.

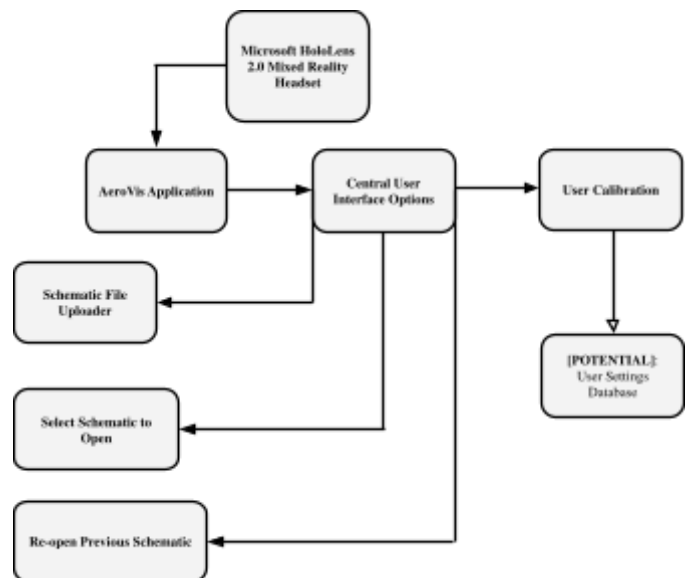


Figure 1: Box diagram displaying the initial foundation and plans for the project direction.

G. Team Charter

Team Charter	
Mixed Reality (MR) Aircraft Wiring Harness Assembly	
Purpose	- This project's purpose is to integrate Mixed Reality (MR) technology into the aircraft wiring harness assembly process to enhance efficiency, accuracy, and reliability.
Goals	- Develop a hardware and software interface for compatibility between existing assembly tools. - Ensure high accuracy and reliability of MR overlays in varying conditions. - Optimize the layout and routing of wiring harnesses to minimize material usage and assembly time.
Team Members and Roles	- Rudy Ferrera: Develop hardware and software interfaces and ensure compatibility with MR technology. - Ryan Monast: Develop hardware and software interfaces and ensure compatibility with MR technology.
Timeline and Milestones	- Phase 1: Research and Planning - Identify existing tools and techniques and assess MR technology requirements.
Meeting Schedule	- Team Time: Every Thursday at 5:30 PM. - Mentor Meetings: Every Tuesday at 11:00 AM.

Figure 2: The team charter used in outlining roles and goals for the team during the duration of this project.

H. Meeting Minutes

Week Two

- ❖ **Time:** 5:30 pm
- ❖ **Discussion:** Discussed project ideas in-depth, exploring their workability, and delving into the intricacies, including electrical and physical components, and software requirements.
- ❖ **Attendees:** Rudy and Ryan
- ❖ **Means of Communication:** In person at the IDEA - Lab.
- ❖ **Advancements Made:** Shared ideas and narrowed down potential project routes with our mentors.
- ❖ **Goals for Next Week:** Set a specific project route and start prototype design.
- ❖ **Other:** Met with the librarian and began our literature search.

Week Three

- ❖ **Time:** 11:30am
- ❖ **Discussion:** Our discussion this week centered around planning the upcoming semester and making sure our goals were not only attainable but reasonable. This discussion also included our mentors.
- ❖ **Attendees:** Ryan, Rudy, Michael, and Ryan
- ❖ **Means of Communication:** Met in person.
- ❖ **Advancements Made:** We have set definitive landmark dates for throughout the semester, and after meeting with our mentors have plans for when we want to meet during the rest of the semester.
- ❖ **Goals for Next Week:** Gantt Chart, Progress Report #1 Completed, Progress Report Presentation #1 Completed, and a functional HUD to demo for the mentors next week.
- ❖ **Other:** We currently sit one week ahead of our schedule and hope to use this momentum to succeed early on.

Week Four

- ❖ **Time:** September 17, 2024 at 11:00 - 11:30 and September 19, 2024 at 4:45 - 6:00
 - ❖ **Discussion:** First, we demoed the progress we had on the hand menus that we made in Unity. We then discussed the issues we ran into during the previous week, and what we wanted to have done for this upcoming week, which we decided was a resizable canvas that the user can interact with.
 - ❖ **Attendees:** Ryan, Rudy, Michael, and Kate
 - ❖ **Means of Communication:** Communicated over a conference call with Michael and Kate, and we discussed via text and email the types of scripts we needed to have functional for the next meeting.
 - ❖ **Advancements Made:** We now successfully have a working demo of what we want our product to look like.
 - ❖ **Goals for Next Week:** We would like to demo a resizable canvas that the user can interact with.
 - ❖ **Other:** Made and discussed our Gantt chart, which we are currently multiple weeks ahead of schedule with.
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Week Five

- ❖ **Time:** September 25, 2024 from 1:30 to 2:30
- ❖ **Discussion:** Since we did not meet with our mentors this week, we discussed the physical demonstration idea we wanted to present the team with during next week's meeting so we can get started on it ASAP.
- ❖ **Attendees:** Ryan and Rudy
- ❖ **Means of Communication:** In-Person and via text message.
- ❖ **Advancements Made:** We have a second demonstration prepared on the development progress made during the previous week, and we have prepared a pitch for the physical demonstration to present next week.
- ❖ **Goals for Next Week:** Get the go ahead to start work on the physical demonstration and work on following the ChangeLogs we have posted [here](#).
- ❖ **Other:** Ryan has taken to documenting a large majority of the project directly to GitHub and is keeping close track over any and all changes week by week.

Week Six

- ❖ **Time:** Thursday 5:30
- ❖ **Discussion:** This week we discussed changes to the UI and started working on the beginning of the physical demonstration.
- ❖ **Attendees:** Rudy, Ryan, and Michael.
- ❖ **Means of Communication:** Microsoft teams meeting.
- ❖ **Advancements Made:** We made significant progress in the blueprints for our physical demonstration and swapped all of the UI elements to the new ideas presented by Michael.
- ❖ **Goals for Next Week:** Have progress made in the physical demonstration and transition fully to the new UI.
- ❖ **Other:** N/A

Week Seven

- ❖ **Time:** Thursday October 15, 2024
- ❖ **Discussion:** Physical demonstration progress and coming plans for the software.
- ❖ **Attendees:** Ryan, Ryan, and Kate
- ❖ **Means of Communication:** Microsoft Teams meeting.
- ❖ **Advancements Made:** Plans for the next week and catching the mentors up on the progress we have made.
- ❖ **Goals for Next Week:** Start ordering parts for the physical demonstration and work with one another to get as much work done on it as possible.
- ❖ **Other:** Since part of this week was during Fall Break less progress than normal was made, but it is alright we are still in a good place.

Week Eight

- ❖ **Time:** Monday October 21st 10:45 - 8:00 and Tuesday 11:00 - 11:30.
 - ❖ **Discussion:** No discussions regarding the project, only work was done.
 - ❖ **Attendees:** Rudy and Ryan
 - ❖ **Means of Communication:** In person communication.
 - ❖ **Advancements Made:** Physical design progress and discussion.
 - ❖ **Goals for Next Week:** Progress Report #2.
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Week Nine

- ❖ **Time:** Tuesday 11:00 - 11:30
- ❖ **Discussion:** Discussion centered around making the physical demonstration and presenting my ideas to the mentors.
- ❖ **Attendees:** Ryan, Ryan, and Kate.
- ❖ **Means of Communication:** Microsoft Teams
- ❖ **Advancements Made:** Lots of information about Vuforia and integrating it into the project.
- ❖ **Goals for Next Week:** Have Vuforia integrated into the project without bugs.

Week Ten

- ❖ **Time:** Tuesday 11:00 - 11:30
- ❖ **Discussion:** Progress report presentation and physical demonstration review.
- ❖ **Attendees:** Rudy, Ryan, Kate, and Michael
- ❖ **Means of Communication:** Microsoft Teams meeting
- ❖ **Advancements Made:** Kate was able to answer some key questions regarding where to find various resources for the implementation of Vuforia Engine into our Unity Scene.
- ❖ **Goals for Next Week:** Make as much progress on the barcode scanner as possible in the engine.

Week Eleven

SIKORSKY VISIT, NO MEETING MINUTES HELD.

Week Twelve

- ❖ **Time:** Tuesday 11:00 - 11:30
 - ❖ **Discussion:** How the specifics of workplace integration would work in terms of our project. Showcase of Vuforia progress and basic A* functionality.
 - ❖ **Attendees:** Ryan, Rudy, Michael, Kate, and Ryan
 - ❖ **Means of Communication:** Microsoft Teams Meeting.
 - ❖ **Advancements Made:** Pivotal questions answered, progress displayed.
 - ❖ **Goals for Next Week:** Advance functionality of the A* algorithm for the next meeting.
 - ❖ **Other:** MASSIVE progress. Great week.
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I. Project Constraints

AeroVis also made sure to take into account many of the constraints societally, economically, and politically may be faced by the product during the development or implementation. For starters, the use of Mixed Reality promotes the opportunity of growth post-release for AeroVis due to the novelty of the technology and the exponential growth of the AR/VR/MR field at this time. Another important constraint AeroVis faces is the price tag. Depending on the organization, there is a price tag of \$3,500 for a new HoloLens2, this price could either be seen as reasonable for larger businesses; however, smaller businesses might not be able to spare such an expense at this time. It is currently hard to tell the projection of a product such as the HoloLens due to Microsoft halting production, however, other devices such as the Meta Quest and the Apple Vision Pro are still being produced. In terms of the user, AeroVis provides a substantial benefit over the traditional method promoting the ability to move around more during the manufacturing process, exposure to significantly less blue light than a traditional screen, and promotes more locational flexibility for the user. Currently, there are no political policies in place that could hurt the product, nor are there expected to be any in the coming future.

J. First Semester Progression

The first weeks of the project were spent studying the desired functions, design specifications, and previous foundations from the group before us. In the third week of the project, the group made the first alpha version of the program available on GitHub for the project. This first version included very rudimentary features including basic hand tracking and early control layouts. This milestone was also monumental because the program was able to be successfully uploaded and tested on the HoloLens, which was a concern for the group last year. Building off of this application, within a week the project had nearly doubled in scope. This was in large part due to the material in the first three reference sources containing similar content to what was being worked on at that time. This second version was now equipped with a fully functional hand menu and schematic visualization. However, the next few weeks were spent bug fixing the problems from the second version. These bugs included head tracking issues, schematic and hand menu clipping errors, and LiDAR issues with the canvas display. During the debugging process it was also important to the project to have a physical demonstration, since only one person can be in the HoloLens at once. The group decided that Rudy, using his experience in C coding,

3D printing, and arduino development, would work primarily on the physical demonstration. This in turn meant Ryan would be doing the work on the software side of the project, since he knows more about C++, Unity, and had a better computer to handle the expensive rendering requirements for an application such as AeroVis. In the more recent weeks, seeing the previous successes in the earlier versions, the mentors of the group thought it would be an amazing idea to implement a barcode scanner, which would allow the user to not have to move the schematic one step at a time. While Ryan worked to implement this idea using the recommended program, Vuforia, Rudy caught the demonstration up to where the Unity software is just in time for the third release. This release included early Vuforia editions, button labeling updates, and canvas updates.

K. Looking Forward

The next steps in the project will be vital in the overall completion of our goal. This is because in the coming semester the project will largely shift to implementing the bulk of the scripting, since a majority of the work before this point has been UI focused. The physical demonstration will now take a back seat as the Unity application looks to move into a Beta phase.

II. REFERENCES

- [1] E. B. Aydoğan and Y. Atay, "Unity Based A* Algorithm Used in Shortest Path Finding Problem for Helicopters," 2021 International Conference on Control, Automation and Diagnosis (ICCAD), Grenoble, France, 2021, pp. 1-5, doi: 10.1109/ICCAD52417.2021.9638762.
 - [2] M. O. Denisov, I. N. Yakovina and I. O. Burov, "Using Mixed Reality for Development of Instruction for Industrial Unit Maintenance," 2023 IEEE 24th International Conference of Young Professionals in Electron Devices and Materials (EDM), Novosibirsk, Russian Federation, 2023, pp. 1850-1854, doi: 10.1109/EDM58354.2023.10225159.
 - [3] A. Evlampev and M. Ostanin, "Obstacle avoidance for robotic manipulator using Mixed reality glasses," 2019 3rd School on Dynamics of Complex Networks and their Application in Intellectual Robotics (DCNAIR), Innopolis, Russia, 2019, pp. 46-48, doi: 10.1109/DCNAIR.2019.8875555.
 - [4] Y. Li, Y. Hu, Z. Wang and X. Shen, "Evaluating the Object-Centered User Interface in Head-Worn Mixed Reality Environment," 2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Singapore, Singapore, 2022, pp. 414-421, doi: 10.1109/ISMAR55827.2022.00057.
 - [5] N. M. Borgest, S. A. Vlasov and N. A. Kondrashin, "Creating an Application with Mixed Reality Technology in the Field of Aircraft Engineering," 2024 International Russian Smart Industry Conference (SmartIndustryCon), Sochi, Russian Federation, 2024, pp. 433-437, doi: 10.1109/SmartIndustryCon61328.2024.10515519.
 - [6] W. Zhang and J. Chen, "Research on Experiential Design of RV Interior Using HoloLens2," 2020 International Conference on Innovation Design and Digital Technology (ICIDDT), Zhenjing, China, 2020, pp. 202-206, doi: 10.1109/ICIDDT52279.2020.00044.
 - [7] K. Nakamura and K. Sekiyama, "Robot Symbolic Motion Planning and Task Execution Based on Mixed Reality Operation," in IEEE Access, vol. 11, pp. 112753-112763, 2023, doi: 10.1109/ACCESS.2023.3322933.
 - [8] Y. Yang and Y. Du, "Application of Improved A* Algorithm in Game Pathfinding," 2024 4th International Conference on Consumer Electronics and Computer Engineering (ICCECE), Guangzhou, China, 2024, pp. 498-501, doi: 10.1109/ICCECE61317.2024.10504153.
 - [9] Y. Sazaki, H. Satria and M. Syahroyni, "Comparison of A* and dynamic pathfinding algorithm with dynamic pathfinding algorithm for NPC on car racing game," 2017 11th International Conference on Telecommunication Systems Services and Applications (TSSA), Lombok, Indonesia, 2017, pp. 1-6, doi: 10.1109/TSSA.2017.8272918.
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